

An Efficient Joint Learning Approach for Item Response Theory

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Introduction

Item Response Theory (IRT) models binary responses from user with latent ability θ and item with latent difficulty β .

Applications: educational testing (e.g., GRE/GMAT), psychometrics, and increasingly, *evaluation of LLMs*.

The most popular model for IRT is the **Rasch model**, where the binary response X_{li} of user l with latent ability θ_l to item i with latent difficulty β_i is

$$\Pr(X_{li} = 1) = \frac{1}{1 + \exp(-(\theta_l - \beta_i))}$$

Goal: Recover both θ and β from binary response matrix.

Notation

n users, m items

$\phi = (\{\theta_l\}_{l=1}^n, \{\beta_i\}_{i=1}^m)$

$\mathcal{O}$: Index set of observed entries

$\eta_{li} = \theta_l - \beta_i$

Note: Rasch model is invariant to additive scaling, hence we impose the centering constraint $\sum_i \beta_i = 0$ to ensure identifiability.

Where prior methods fall short

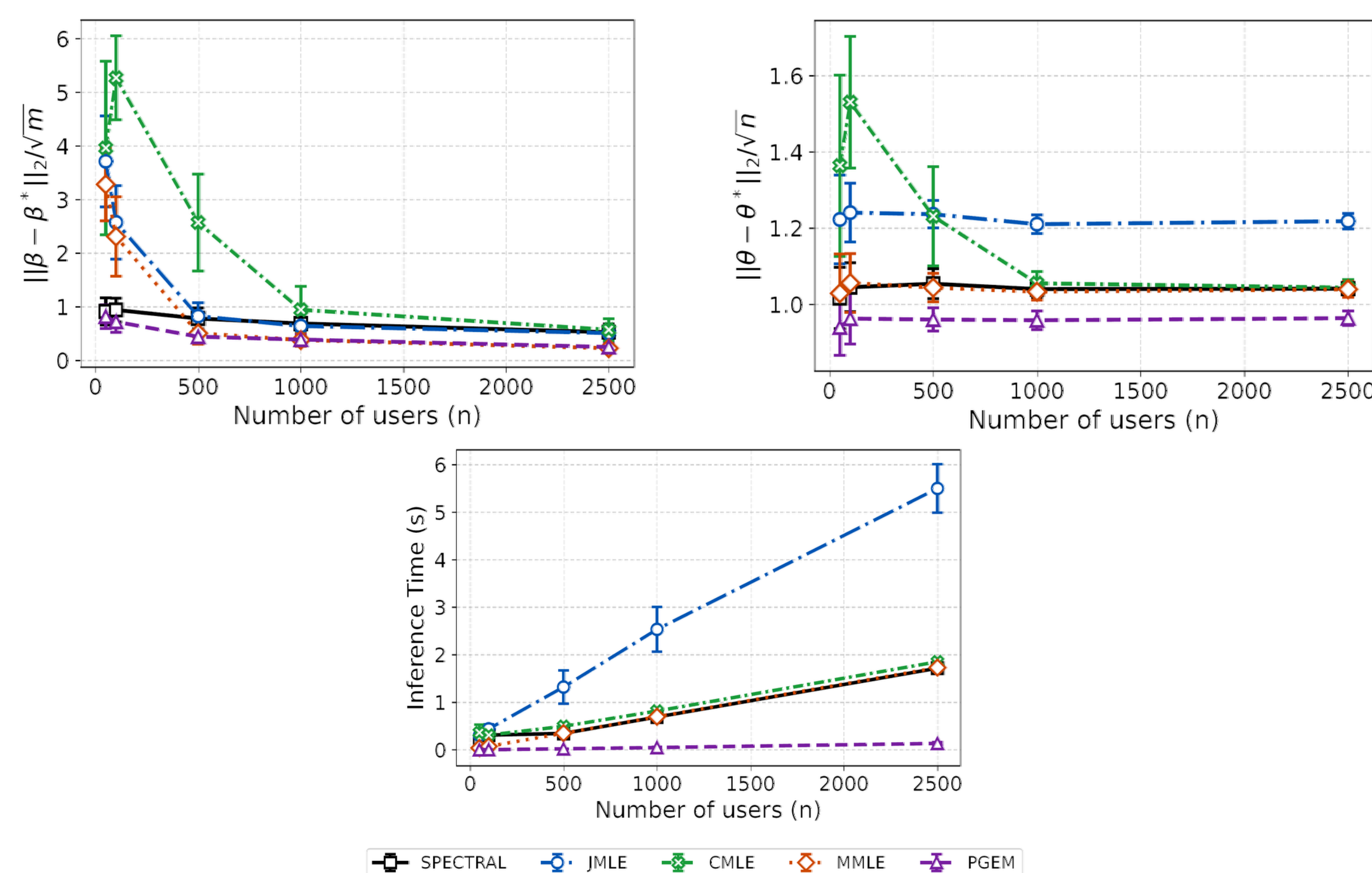
Joint Maximum Likelihood Estimation (JMLE)
known to produce inconsistent estimates [1]

Marginal Maximum Likelihood Estimation (MMLE)
prone to prior misspecification

Conditional Maximum Likelihood Estimation (CMLE)
computationally expensive

Spectral Approach [2]
requires dense regimes; computationally expensive in larger datasets

All above methods other than JMLE estimate only item parameters



Synthetic Experiments: $m = 10$, density $(p) = 0.05$, varying the number of users. We can observe how the existing methods struggle.

Polya-Gamma Augmentation

$$\mathcal{L}_{\text{obs}}(\phi; X) \propto \prod_{(l,i) \in \mathcal{O}} \frac{1}{1 + \exp\{(-1)^{X_{li}}(\theta_l - \beta_i)\}}$$

Observed-data likelihood under Rasch model

$$\frac{1}{1 + e^\eta} = \frac{1}{2} e^{-\eta/2} \int_0^\infty e^{-\omega \eta^2/2} p(\omega) d\omega$$

Identity from [3]: Introducing a latent Polya-Gamma (PG) random variable ω for each observation turns the logistic form into Gaussian

$$p(X_{li}, \omega_{li} | \phi) \propto \exp\left(-\frac{(-1)^{X_{li}} \eta_{li}}{2} - \frac{\omega_{li} \eta_{li}^2}{2}\right) p(\omega_{li})$$

Augmented joint density

$$\ell(\phi) = -\frac{1}{|\mathcal{O}|} \sum_{(l,i) \in \mathcal{O}} \left[\frac{(-1)^{X_{li}}}{2} (\theta_l - \beta_i) + \frac{1}{2} \omega_{li} (\theta_l - \beta_i)^2 \right]$$

Quadratic log-likelihood (given all ω)

PGEM Algorithm

Initialize $\phi^{(0)}$ randomly.

E-step

PG distribution moments lead to a *closed form* expectation over ω :

$$Q(\phi | \phi^{(t)}) = -\frac{1}{|\mathcal{O}|} \sum_{(l,i) \in \mathcal{O}} \left[\frac{(-1)^{X_{li}}}{2} (\theta_l - \beta_i) + \frac{1}{2} \kappa_{li}^{(t)} (\theta_l - \beta_i)^2 \right]$$

$$\text{where } \kappa_{li}^{(t)} = \frac{1}{2 \eta_{li}^{(t)}} \tanh\left(\frac{\eta_{li}^{(t)}}{2}\right)$$

M-step

Maximize regularized Q-function. The ℓ_2 regularization keeps estimates well-posed if MLE diverges

$$\phi^{(t+1)} \leftarrow \arg \max_{\phi} Q(\phi | \phi^{(t)}) - \frac{\nu}{2} \|\phi\|_2^2.$$

Equivalent to solving a linear system:

$$H\phi = r$$

Under centering constraint, H is positive definite and strictly diagonally dominant. We use Gauss-Seidel method to solve the linear system.

Repeat until convergence.

Theoretical Guarantees

Building on [4], Let $\phi^* = \arg \max_{\phi} \tilde{Q}(\phi | \phi^*)$ and let $\mathcal{B}_2(r; \phi^*)$ denote the Euclidean ball of radius r centered at ϕ^* .

Assumptions:

- Responses in X are observed uniformly at random
- Regularization parameter $\nu \geq 0.5$

Linear Convergence: For any initial point $\phi^{(0)} \in \mathcal{B}_2(r; \phi^*)$,

$$\|\phi^{(t)} - \phi^*\|_2 \leq \rho^t \|\phi^{(0)} - \phi^*\|_2, \quad \rho \leq 0.8$$

Finite-Sample Bound: If the sample size

$$N \geq \frac{6K}{(1-\rho)r} \cdot \max\{n, m\} \cdot \ln\left(\frac{4 \min\{n, m\}}{\delta}\right)$$

for a sufficiently large constant K , then with probability at least $1-\delta$,

$$\|\phi^{(t)} - \phi^*\|_2 \leq \rho^t \|\phi^{(0)} - \phi^*\|_2 + \frac{6K}{1-\rho} \cdot \frac{\max\{n, m\} \ln\left(\frac{4 \min\{n, m\}}{\delta}\right)}{N}$$

Real-Data Experiments

Table 1. Metric-wise comparison across original datasets. Best values are highlighted per dataset. Higher is better for AUC, LogLik, and Accuracy; lower is better for Time and Brier.

Method	ICAR-SAPA					MTurk-Lexical					ML-100K				
	Time	AUC	LogLik	Acc	Brier	Time	AUC	LogLik	Acc	Brier	Time	AUC	LogLik	Acc	Brier
JMLE	210.79	0.785	-0.598	0.719	0.190	33.42	0.800	-0.328	0.889	0.087	32.47	0.726	-0.655	0.671	0.219
MMLE	845.40	0.790	-0.604	0.721	0.192	51.57	0.784	-0.383	0.883	0.092	22.96	0.730	-0.627	0.675	0.211
CMLE	> 25k			NA		6953.60	0.550	-0.698	0.829	0.139	4407.60	0.475	-1.260	0.477	0.377
SPECTRAL	2128.68	0.804	-0.538	0.724	0.181	3.49	0.811	-0.295	0.885	0.087	2.86	0.728	-0.607	0.670	0.210
PGEM	52.71	0.814	-0.524	0.734	0.176	3.46	0.831	-0.279	0.891	0.082	1.18	0.739	-0.598	0.681	0.205

Method	HETREC					BOOK-GENOME					ML-20M				
	Time	AUC	LogLik	Acc	Brier	Time	AUC	LogLik	Acc	Brier	Time	AUC	LogLik	Acc	Brier
JMLE	139.74	0.746	-0.623	0.691	0.208	31566.2	0.657	-0.838	0.615	0.263	26191.6	0.724	-0.664	0.670	0.221
MMLE	122.85	0.754	-0.597	0.697	0.200	1509.21	0.688	-0.638	0.641	0.223	1522.63	0.746	-0.594	0.688	0.203
CMLE	> 15k			NA					NA					NA	
SPECTRAL	111.96	0.752	-0.588	0.695	0.201	10630	0.676	-0.644	0.633	0.226	53174.9	0.745	-0.592	0.687	0.203
PGEM	24.69	0.759	-0.578	0.700	0.197	8318.86	0.698	-0.634	0.647	0.221	4223.38	0.747	-0.590	0.689	0.202

References

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- [3] Polson, N. G., Scott, J. G., and Windle, J. Bayesian inference for logistic models using p’olya-gamma latent variables. Journal of the American Statistical Association, 108(504):1339–1349, 2013. doi: 10.1080/01621459.2013.829001.
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